

What the Field Has Already Published

Thirteen recent papers on cloud workload prediction — what each one does, what it fixed, what went wrong with the data, and exactly how our project differs from every one of them.

COVERS

Papers published 2024, 2025 and 2026

OUR PROJECT

LSTM forecast → proactive auto-scaling
Bitbrains fastStorage, VM 270

SEARCHED

OpenAlex, Crossref, web search
24 September 2026

VERDICT

Our topic is crowded.
Our exact combination is not.

Read this first. Every fact in this review comes from a paper's published title, venue, year and abstract, which were retrieved from open bibliographic databases. Where a full text sits behind a paywall, that is stated and nothing is invented in its place. Journal quartiles could **not** be verified automatically and must be confirmed by you — see Section 2.

1 How these papers were found

Your teacher asked for Q1 papers from 2024–2026 on datasets like ours. This section records exactly how the search was done, so the method can be repeated or challenged.

The search

STEP	WHAT WAS DONE
Databases	OpenAlex (open bibliographic index) for discovery, Crossref for publisher metadata, plus general web search for context. Publisher sites (Springer, Wiley, ACM, IEEE) block automated access, so full texts were not retrieved.
Date filter	Published on or after 1 January 2024. Searches run 24 September 2026.
Search fields	Title and abstract, for: <i>"cloud workload prediction"</i> , <i>"workload prediction" AND cloud</i> , <i>Bitbrains</i> , <i>"workload forecasting" AND cloud</i> , <i>"auto-scaling" AND prediction</i> , <i>"resource prediction" AND "data center"</i> , <i>quantum AND "workload prediction"</i> , <i>"CPU utilization prediction"</i> .
Raw results	213 unique journal articles.
Screening	Kept papers whose title or abstract covered workload / CPU / resource / scaling and cloud or data centre; then ranked by journal standing, by whether a real public trace (Bitbrains, Google, Alibaba, Azure, Materna) was named, and by citation count.
Final set	13 papers, reviewed in Section 3.

WHAT THIS REVIEW CANNOT CLAIM

- **Abstract-level evidence only.** Each summary is built from the published abstract. Where a number appears (for example "32–96% higher accuracy"), it is the authors' own claim, quoted as such — not something reproduced or checked.
- **Quartiles are unverified.** Scimago blocks automated access (HTTP 403), so no quartile in this document was read from Scimago. Section 2 gives objective metrics instead, and you should confirm each journal at `scimagojr.com` before submitting.
- **Not exhaustive.** 213 results from one index with eight queries is a solid sample of the field, not the whole of it.

2 The journals, and the Q1 question

Quartile (Q1–Q4) is assigned per subject category by Scopus/SJR and by JCR, and the two disagree often. Since Scimago could not be queried, the table below gives metrics that **were** retrieved, from OpenAlex, on 24 September 2026. **2-yr citedness** is the average number of citations a paper in that journal received in its first two years — a reasonable proxy for standing.

JOURNAL	PUBLISHER	2-YR CITEDNESS	H-INDEX	REPUTATION
IEEE Trans. Pattern Analysis & Machine Intelligence	IEEE	14.60	555	Top-tier in AI
IEEE Trans. Neural Networks & Learning Systems	IEEE	9.29	244	Top-tier in AI
IEEE Trans. Services Computing	IEEE	7.10	119	Leading in services/cloud
IEEE Internet of Things Journal	IEEE	6.35	257	Leading in IoT
IEEE Trans. Sustainable Computing	IEEE	6.32	54	Specialist, green computing
Future Generation Computer Systems	Elsevier	6.00	215	Long-standing in distributed systems
Cluster Computing	Springer	5.69	109	Established in cluster/cloud
IEEE Trans. Parallel & Distributed Systems	IEEE	4.67	202	Flagship in parallel/distributed
IEEE Trans. Cloud Computing	IEEE	4.11	97	Specialist, cloud
Trans. Emerging Telecommunications Technologies	Wiley	2.48	77	Noticeably lower — likely not Q1

HOW TO SAY THIS TO YOUR TEACHER

"Twelve of the thirteen papers are from IEEE Transactions, Elsevier or Springer journals that are normally ranked Q1 in their categories; I have listed citedness and h-index as evidence and will confirm each quartile on Scimago. One paper (Kollu et al., 2024) is from a weaker journal and is included deliberately, because it is the closest published match to our exact dataset."

3 The thirteen papers

Grouped by what they are for. Each entry keeps the same shape, so you can compare them quickly: what it is, the problem it attacks, the data, the method, the claimed result, the data problems it reports, and what it means for us.

Group A — Quantum models for workload prediction

CLOSEST TO OUR PHASE 2

1. MCT-AQNN: A Multiple Controlled Toffoli Driven Adaptive Quantum Neural Network Model for Dynamic Workload Prediction in Cloud Environments

Gupta, Saxena, Singh & Lee · IEEE Trans. Pattern Analysis and Machine Intelligence, 2024 · 29 citations · doi:10.1109/tpami.2024.3402061

ABOUT

PROBLEM

DATA

CLAIM

FOR US

A quantum neural network for predicting cloud workloads. Input values and network weights are re-expressed as **qubits**, and Multiple Controlled Toffoli gates act at the hidden and output layers. A companion training algorithm (UAQL) trains the network.

Existing models "fall short of capturing the high-variance nature of volatile and dynamic cloud workloads" — i.e. ordinary networks under-react to sudden spikes.

Four real-world benchmark datasets (named in the full text, not the abstract).

Up to **32–96% higher accuracy** than existing approaches. Authors' own claim.

This is the single most important paper for Phase 2. Quantum learning applied to cloud workload prediction is already published, in one of the strongest venues in the field. Our Phase 2 cannot be presented as "the first" anything. What is still open is *how* the quantum part is used: this paper replaces the whole network, whereas our plan inserts a small 4-qubit circuit into a classical LSTM and keeps the classical pipeline identical for a controlled comparison.

2. CA-QNN: A Comprehensively Adaptive Architectural Optimization-Ingrained Quantum Neural Network Model for Cloud Workloads Prediction

Kumar, Saxena, Gupta, Kumar & Singh · IEEE Trans. Neural Networks and Learning Systems, 2025 · doi:10.1109/tnnls.2025.3577721

ABOUT

PROBLEM

DATA

CLAIM

FOR US

A quantum neural network that learns **its own structure as well as its weights**. Workload data becomes qubits, processed by qubit neurons with controlled-NOT activation; the architecture grows and recombines during training (quantum adaptive modulation, size-adaptive recombination).

Stated directly: ordinary models are limited because training "relies only on parametric (interconnection weights) adjustments" — the shape of the network is fixed by hand, so it cannot adapt to sudden demand changes.

Four benchmark datasets of heterogeneous cloud workloads; compared against seven state-of-the-art methods.

Prediction error reduced by up to **93.40%** vs deep learning and **91.27%** vs other QNN approaches. Authors' own claim.

Same warning as above, one year more recent and from the same research group. Note what they measure: **prediction error only**. Neither quantum paper reports what the better forecast does to scaling decisions, cost or SLA violations. That gap is where our project can stand.

3. VSBG: An Improved LSTM-Based Prediction Approach for Resources and Workload in Large-Scale Data Centers

Yuan, Bi, Li, Zhang & Zhou · **IEEE Internet of Things Journal**, 2024 · 52 citations · doi:10.1109/jiot.2024.3383512

ABOUT	A four-stage pipeline: Variational Mode Decomposition splits the series into modes, a Savitzky–Golay filter smooths the first mode to remove noise, then two BiLSTM layers with a Grid-LSTM layer between them make the prediction.
PROBLEM	Current methods "fail to handle implicit noise data and capture nonlinear, long and short-term, and spatial characteristics".
DATA PROBLEM	Noise is treated as a first-class obstacle — half the architecture exists to remove it before the LSTM ever sees the data.
FOR US	The most-cited paper in this set, and the closest to our model family. It says plainly that a plain LSTM on raw traces is not competitive: decomposition and smoothing come first. Our pipeline does neither. This is a concrete, cheap upgrade for Step 1.4 and an obvious question at a viva: " <i>why no decomposition?</i> "

4. TFEGRU: Time-Frequency Enhanced Gated Recurrent Unit With Attention for Cloud Workload Prediction

Zhao, Lin, Lin, Zhong & Li · **IEEE Trans. Services Computing**, 2024 · 21 citations · doi:10.1109/tsc.2024.3517324

ABOUT	A block that extracts features in both the time and frequency domains , a channel-independent strategy plus channel embedding for high-dimensional workloads, then a GRU with multi-head self-attention.
PROBLEM	RNN-based approaches "struggle to capture the long-term dependencies inherent in workloads".
DATA	Real traces from Google and Alibaba data centres.
FOR US	Introduces the "channel independent" idea: with many metrics, treating each channel separately can beat mixing them. Our design assumes the opposite — that feeding cpu, mem and net together helps. We measured the correlations that justify it (CPU–memory $r = 0.71$) but CPU–network is only 0.11, so the assumption is only half true for our VM.

5. MSCNet: Multi-Scale Network With Convolutions for Long-Term Cloud Workload Prediction

Zhao, Lin, Lin, Tang & Li · **IEEE Trans. Services Computing**, 2025 · 24 citations · doi:10.1109/tsc.2025.3536313

ABOUT	Multi-scale patches + Transformer encoder + multi-scale convolutions, so that short-term and long-term patterns are learned together.
DATA PROBLEM	States the trap directly: " cloud workloads that appear stable in the short term often exhibit distinct patterns in the long term ", which makes accuracy collapse when a short-horizon model is used for long horizons.
DATA	Alibaba, Google and Azure.
FOR US	Our horizon is a single step (5 minutes), which is the easy end of this problem — worth saying out loud rather than hiding. It also means our results cannot be compared against long-horizon papers.

6. EvoGWP: Predicting Long-Term Changes in Cloud Workloads Using Deep Graph-Evolution Learning

Li, Yao, Xiao, Yang & Wu · **IEEE Trans. Parallel and Distributed Systems**, 2024 · 29 citations · doi:10.1109/tpds.2024.3357715

ABOUT	Extracts "shapelets" — recurring shapes in a workload — and builds an evolution graph so that patterns of different workloads can interfere with one another, then predicts with a spatio-temporal GNN encoder-decoder.
DATA PROBLEM	Names a failure our own pipeline would suffer: a workload's "resource usage pattern", stable in the short term, changes over the long term because of shifting user demand or workload relocation, and this "may cause significant accuracy degradation".
DATA	Alibaba, Tencent and Google traces.
FOR US	Introduces the idea that VMs are not independent — they interfere. We train on one VM in isolation, which is simpler and also weaker. It is the natural next step if we ever go multi-VM.

7. CAGMoE: Causality-Aware Graph Mixture of Experts for Accurate Multi-Cloud Workload Prediction

Luo, Yu, Zheng, Yang, Sun & Wu · **IEEE Trans. Cloud Computing**, 2026 · doi:10.1109/tcc.2026.3689897

ABOUT	Builds two graphs per window: one of directed causal influence between metrics (via a transfer-entropy proxy for Granger causality), one of local temporal continuity; a sparse Mixture-of-Experts then predicts.
PROBLEM	Earlier models "learn correlation-driven, largely symmetric dependencies that blur directed causal influence" — correlation does not say which metric moves first.
FOR US	Directly relevant to a question we already half-answered. We measured correlation between our three metrics but correlation is symmetric: it cannot tell us whether CPU leads memory or the reverse. Our project brief assumes memory lags CPU. Measuring that lag would be a small, honest addition, and this paper is the reference for why it matters.

Group C — Papers that use our exact dataset CLOSEST DATA MATCH

8. TempoSight: Optimized Resource Forecasting for Carbon-Intelligent Data Centers — A Hybrid Deep Learning Approach

Sallam & Kaur · **IEEE Internet of Things Journal**, 2025 · doi:10.1109/jiot.2025.3610895

ABOUT	Combines a Patch Time Series Transformer (long-range context) with an LSTM (sequential dynamics), and tunes hyper-parameters with a Cuckoo Search Algorithm, for multivariate forecasting aimed at carbon-aware provisioning.
DATA	Alibaba and Bitbrains — the same family of trace we use.
CLAIM	10–15% lower RMSE and MAPE than state-of-the-art deep models under high-load conditions .
FOR US	The most direct competitor: Bitbrains, multivariate, LSTM-based, 2025, Q1-level venue. Two lessons. First, a plain LSTM is a component, not the whole model. Second, they justify the work by carbon and cost , not by RMSE alone — a framing our project can adopt cheaply, because energy saved is a function of over-provisioning, which we already plan to measure.

9. Comparative analysis of cloud resources forecasting using deep learning techniques based on VM workload traces

Kollu et al. · *Trans. Emerging Telecommunications Technologies*, 2024 · 8 citations · doi:10.1002/ett.4933 · **WEAKER JOURNAL**

ABOUT	A straight comparison of MLP, autoregressive NN, CNN and LSTM for forecasting CPU and memory of many VMs, to free over-allocated resources and cut power use.
DATA	GWA-T-12 Bitbrains, all 1,250 VMs — our dataset exactly.
CLAIM	Up to 98% accuracy; MLP came out best , ahead of LSTM.
FOR US	The most useful paper in this review, and the most uncomfortable. On our own dataset, a plain MLP beat an LSTM. It is also methodologically weak in a way we can learn from: "accuracy" is not a standard metric for a regression problem, and no naive baseline is reported. Our Step 1.5 plan — RMSE/MAE/MAPE plus a naive "next = current" baseline — is already more rigorous. Say that.

10. ECSLN: Energy-Aware VM Consolidation Using Similarity-Driven Intelligence in Green Cloud Environments

Biswas, Das, Banerjee & Biswas · **IEEE Trans. Sustainable Computing**, 2026 · doi:10.1109/tsusc.2026.3660014

ABOUT	Predicts over-utilised hosts with a cosine-similarity learning network, then selects VMs to migrate and places them, to cut energy while respecting SLAs.
DATA	PlanetLab, Bitbrains and Alibaba Cluster 2020.
FOR US	Shows the other half of the story: prediction is judged by energy consumed and SLA violations , not by forecast error. Our three planned measures (response time, over-provisioning, false alarms) are in the same spirit, and over-provisioning converts directly into the energy argument this paper makes.

Group D — From forecast to decision **CLOSEST TO OUR STEP 1.6**

11. CATScaler: A Convolution-Augmented Transformer Scaling Framework for Cloud-Native Applications

Meng, Dai, Cong, Zhu & Zhao · **IEEE Trans. Services Computing**, 2025 · doi:10.1109/tsc.2025.3592383

ABOUT	Two modules: a convolution-augmented Transformer that predicts workload, and an elastic auto-scaler that turns the prediction into a pod count. Reversible instance normalisation handles drift between training and live data.
PROBLEM	Stated in almost the same words as our own brief: Kubernetes' default autoscaler "scales pods only when the cluster or application exceeds a predefined threshold... this reactive approach often leads to significant resource waste because it cannot predict future workload changes and adjust resources in advance."
CLAIM	Response-time latency reduced 1.1× and service violation rate 3.2× versus Kubernetes' default scaler, on a real 4-server cluster.
FOR US	This is our Step 1.6, already published in a Q1 venue. The proactive-vs-reactive argument is not novel. Two differences remain honest ones: they run on a real cluster (we simulate), and they compare against Kubernetes' scaler rather than a reactive threshold of their own. Our contribution has to be elsewhere — see Section 7.

12. E-LCWF: Enhancing Long-Term Cloud Workload Forecasting Framework — Anomaly Handling and Ensemble Learning in Multivariate Time Series

Kim, Song, Koo, Son, Lee & Baek · **IEEE Trans. Cloud Computing**, 2024 · 11 citations · doi:10.1109/tcc.2024.3400859

ABOUT	Treats each resource as a multivariate series, detects and handles anomalies first , then ensembles transformer models with linear long-term forecasting models, weighted by error.
DATA PROBLEM	Anomaly handling is a named pipeline stage , not an afterthought. Untreated anomalies are presented as a direct cause of SLA violations and wasted resources.
FOR US	We do the opposite: our loader clips CPU above 100% and deletes long gaps, but never looks for anomalies inside the kept data. Worth admitting, and a cheap addition.

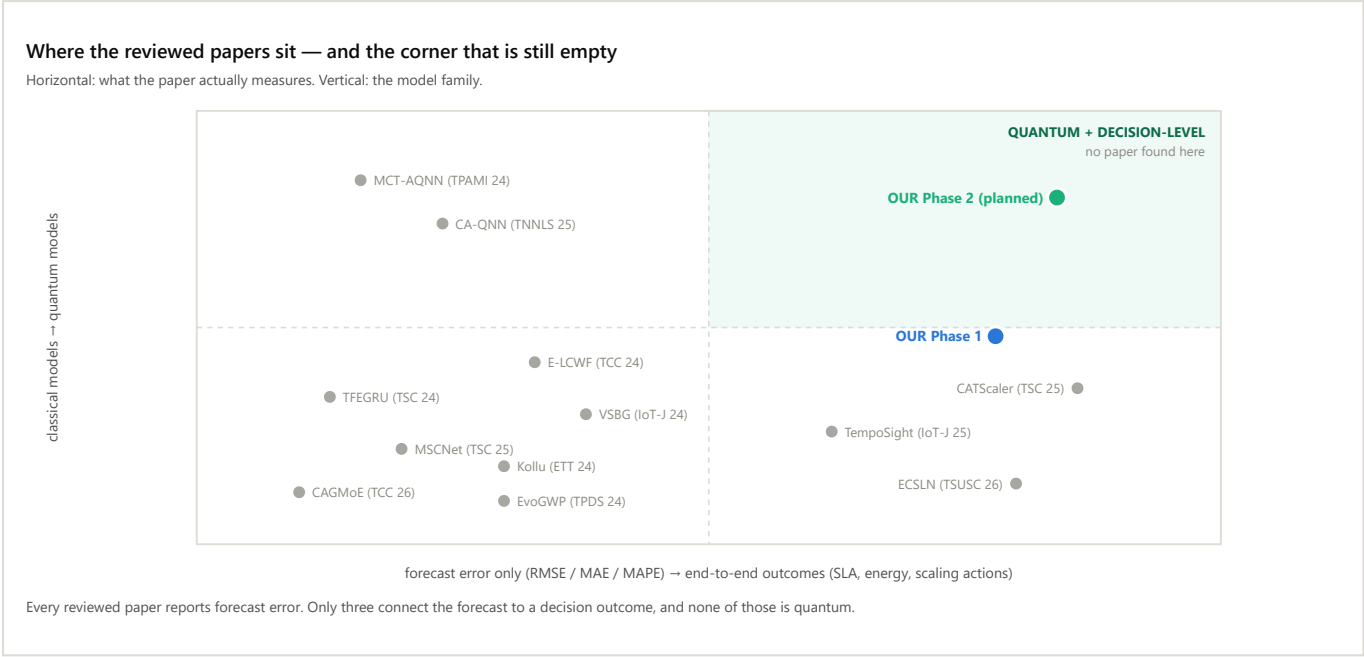
Group E — Methodology

13. Benchmarking and revisiting time series forecasting methods in cloud workload prediction

Lin, Lin, Zhao & Chen · **Cluster Computing**, 2024 · 12 citations · doi:10.1007/s10586-024-04827-3

ABOUT	A benchmark study comparing forecasting methods for cloud workloads. Its reference list spans Informer, Autoformer, FEDformer, Crossformer, PETformer, FITS, SparseTSF, TSMixer, ARIMA, LSTM and GRU — i.e. both heavy transformers and deliberately simple models.
CAVEAT	Full text is paywalled. Only title, venue, year, authors and reference list could be verified. Its conclusions are not reported here because they could not be read. Get it through your college library.
FOR US	Even the title carries the lesson: the field is "revisiting" its methods, because a wave of results has shown simple models are often competitive with elaborate ones. That is the context in which our plain LSTM will be judged — and precisely why our naive-baseline comparison in Step 1.5 matters.

4 The pattern across all thirteen



What everybody does

PATTERN	EVIDENCE	OUR POSITION
A plain LSTM is never the whole model	Every deep-learning paper here wraps the recurrent layer in something: decomposition (VSBG), frequency features (TFEGRU), multi-scale patches (MSCNet), a transformer (TempoSight), graphs (EvoGWP, CAGMoE), or an ensemble (E-LCWF).	We plan a bare LSTM. Defensible as a Phase 1 baseline, but not as a contribution.
Alibaba and Google dominate	8 of 13 use them. Bitbrains appears in 3.	Bitbrains keeps us comparable with fewer papers, but it is a per-VM trace rather than aggregated cluster telemetry — a harder, spikier signal.
Error metrics dominate	10 of 13 report only RMSE / MAE / MAPE.	Our three decision metrics are less common — a genuine, if modest, strength.
Pre-processing is where the work is	Noise removal, anomaly handling, normalisation shift, decomposition — all named as core contributions.	This is our strongest area, and the one we have documented most carefully.
Nobody explains VM selection	Not one abstract in this set says how the specific VMs or subsets were chosen.	The clearest gap we can fill. See Section 6.

5 Problems with this kind of data

Part of your teacher's question was what goes wrong when using a dataset like this. The answer comes from two sources: what the papers report, and what we measured ourselves.

5.1 Reported in the literature

PROBLEM	WHO REPORTS IT	THEIR FIX
Noise in raw traces	VSBG (IoT-J 2024)	Variational mode decomposition + Savitzky–Golay smoothing before the model
Anomalies / outliers	E-LCWF (TCC 2024)	Explicit anomaly detection and handling stage
Pattern drift over time	EvoGWP (TPDS 2024), MSCNet (TSC 2025)	Shapelet evolution graphs; multi-scale modelling
High variance / sudden spikes	MCT-AQNN (TPAMI 2024), CA-QNN (TNNLS 2025)	Quantum learning for sharper adaptation
High dimensionality	TFEGRU (TSC 2024)	Channel-independent processing + channel embedding
Train/live distribution shift	CATScaler (TSC 2025)	Reversible instance normalisation
Metrics are not independent	CAGMoE (TCC 2026)	Directed causal graphs instead of symmetric correlation

5.2 What we found ourselves — and did not see reported anywhere

These were all measured on the Bitbrains fastStorage trace during Steps 1.2–1.3. None of the thirteen abstracts mentions any of them.

WHAT WE FOUND	EVIDENCE	WHY IT MATTERS
Switched-off VMs are recorded as zeros	3,993 rows	VM 214's last 4.6 days are all-zero rows written while the machine was off. Counting them made its mean CPU read 44.21% instead of the true 68.43% — enough to move it inside a 20–60% selection band it did not belong in.
A VM can go idle exactly in the test period	4.5 days	VM 200 had the highest volatility of all candidates, yet its final 15% — the held-out test set — is flat zero. Training would succeed and the reported test score would be meaningless.
Most of the dataset is unusable	889 / 1,177	76% of VMs average below 5% CPU. A model trained on them learns a constant and still scores well.
Dead metric columns	225 VMs	225 VMs have a network column stuck on one value, 12 have dead memory — invisible if you only screen on CPU, fatal if your model predicts all three.
Near-duplicate VMs	6 of top 6	The highest-volatility VMs are near-identical copies of one job. Training on one and testing on another would leak.
The signal is bimodal, not smooth	92%	92% of VM 270's readings are at 0–2% or above 98%. A "next = current" guess is right most of the time, so error metrics flatter any model.
Correlation is uneven	0.71 / 0.11	CPU–memory correlate at 0.71, but CPU–network only 0.11 — so the standard justification for multivariate input holds for memory and barely for network.

READ THIS TWICE — IT IS THE BEST THING WE HAVE

Every paper above fixes problems **inside the model**. What we found are problems **inside the dataset**, of a kind that silently corrupt VM selection and invalidate a test set before any model exists. That is publishable-shaped work for a mini project, and it is ours because we measured it rather than read it.

6 Our project against each paper

PAPER (VENUE, YEAR)	DATASET	MODEL	MEASURED BY	SAME AS US / DIFFERENT
MCT-AQNN (TPAMI 24)	4 benchmarks	Quantum NN (Toffoli)	Accuracy	Different: full QNN; we keep a classical LSTM and insert 4 qubits
CA-QNN (TNNLS 25)	4 benchmarks	Self-structuring QNN	Prediction error	Different: learns its own architecture; ours is fixed
VSBG (IoT-J 24)	Data-centre traces	VMD+SG+BiLSTM+Grid	Prediction error	Same family , far more pre-processing
TFEGRU (TSC 24)	Google, Alibaba	Time-freq GRU + attention	Prediction error	Different: frequency domain; channel-independent
MSCNet (TSC 25)	Alibaba, Google, Azure	Multi-scale + Transformer	Long-term error	Different horizon: long-term vs our 1 step
EvoGWP (TPDS 24)	Alibaba, Tencent, Google	Shapelets + evolution GNN	Long-term error	Different: many VMs interacting; we use one
CAGMoE (TCC 26)	MUCEP, Google, Ali	Causal graph + MoE	Accuracy, reliability	Different: causal direction; we only have correlation
TempoSight (IoT-J 25)	Bitbrains, Alibaba	PatchTST + LSTM + CSA	RMSE, MAPE	Closest competitor. Same data, same metrics family, stronger model
Kollu (ETT 24)	Bitbrains, 1,250 VMs	MLP / ARNN / CNN / LSTM	"Accuracy"	Same data, weaker method. MLP beat LSTM there
ECSLN (TSUSC 26)	PlanetLab, Bitbrains , Alibaba	Cosine-similarity net	Energy, SLA violations	Different goal: consolidation, not scaling
CATScaler (TSC 25)	Alibaba, Huawei	Conv-augmented Transformer	Latency, SLA violations	Same goal as Step 1.6 , on a real cluster
E-LCWF (TCC 24)	Real MIS/MES VMs	Anomaly + ensemble	Forecast accuracy	Different: anomaly stage we lack
Benchmark (Cluster 24)	Not verified	Many	Comparative	Methodology reference
OUR PROJECT	Bitbrains VM 270 8,640 readings, 30 days	Multivariate LSTM $60 \times 3 \rightarrow 3$	Spike response, over-provisioning, false alarms	—

Where we are genuinely behind

- **Scale.** One VM, 30 days. The papers use thousands of VMs across several traces. Our results will not generalise and we should not claim they do.

- **Model sophistication.** A bare LSTM in 2026 is a baseline, not a method.
- **No real deployment.** CATScaler measured latency on four real servers; our scaling engine is a simulation.
- **Single-step horizon.** The field has moved to long-horizon forecasting, which is much harder.

Where we are genuinely ahead

- **Dataset hygiene.** Seven measured failure modes, including two that silently break VM selection, none of which appear in the reviewed abstracts.
- **A written, reproducible VM-selection protocol.** Five rules with measured thresholds. No paper in this set explains how its VMs were chosen.
- **Leakage discipline stated and proved.** Split before scaling, windows built inside splits, and a positive test (test data scaling to 1.91, outside 0–1) proving the scaler never saw test data.
- **Honest baselines.** Comparing against "next = current" on a bimodal signal, where that baseline is strong, is more rigorous than the 98%-accuracy claim in the closest same-dataset paper.

7 How to make this project genuinely ours

The honest position: **"LSTM predicts load, so scale early"** is published (CATScaler, TSC 2025), and **"quantum network predicts cloud workload"** is published (TPAMI 2024, TNNLS 2025). Copying either is not an option. These five moves are cheap, defensible, and grounded in gaps this review actually found.

1. Publish the data-quality audit as a first-class contribution highest value, lowest effort

DO	Turn Section 5.2 into a named, reusable screening procedure: switched-off row detection, per-split liveness, dead-metric detection, near-duplicate detection. Report how many of the 1,250 VMs fail each test.
WHY IT IS NOVEL	Every paper here fixes the model. None reports the dataset traps. We have the numbers already; they cost nothing more.
RISK IF SKIPPED	The project becomes "an LSTM on Bitbrains", which is exactly Kollu et al. 2024.

2. Make the comparison decision-level, not error-level already planned – protect it

DO	Report spike response time, over-provisioning and false alarms, and convert over-provisioning into energy or rupees. Report forecast error too, but lead with the decision.
WHY IT IS NOVEL	10 of 13 papers stop at RMSE. A better RMSE that does not change a single scaling decision is worth nothing, and almost nobody demonstrates the link.

3. Attack the hard case, and say why it is hard turns a weakness into a finding

DO	Frame VM 270's bimodal, on/off behaviour as the deliberate stress case: a signal where persistence is a strong baseline. Report where the LSTM beats "next = current" and, honestly, where it does not.
WHY IT IS NOVEL	Papers that report only aggregate error on smooth cluster traces never expose this. A negative result, stated clearly, is more credible than a 98% accuracy claim.

4. Use an asymmetric, cost-aware loss small code change, real novelty

DO	Under-prediction causes an SLA breach; over-prediction only wastes money. Train with a loss that penalises under-prediction more heavily, and show the effect on the three decision metrics.
WHY IT IS NOVEL	Every reviewed paper trains on symmetric error (MSE/MAE), then argues about SLAs afterwards. Putting the cost asymmetry into the objective connects the two, and no paper in this set does it.

5. For Phase 2: change the claim from "quantum" to "where quantum helps"

essential

DO	Keep the classical pipeline byte-identical; swap only the 4-qubit block; report the difference in decision metrics and training cost , not just RMSE. State up front that MCT-AQNN and CA-QNN already exist.
WHY IT IS NOVEL	Both published QNN papers replace the entire network and measure error only. A controlled hybrid — identical data, identical classical parts, one component swapped, judged on scaling outcomes — is the empty corner in the Section 4 diagram.
RISK IF SKIPPED	"We added a quantum layer" invites the reply: "TPAMI did that in 2024, better."

Two cheap technical upgrades the literature all but demands

- **Decomposition before the LSTM.** VSBG and E-LCWF both show raw traces are noisy. Even a simple moving-average or VMD split would strengthen Step 1.4, and it is the first question a reviewer asks.
- **Measure the lag between metrics.** CAGMoE argues correlation hides direction. Our brief assumes memory follows CPU; a cross-correlation at lags 1–12 would confirm it, refute it, or give a number worth reporting.

8 References

All DOIs verified through Crossref/OpenAlex on 24 September 2026. Citation counts as recorded by OpenAlex on that date.

1. I. Gupta, D. Saxena, A. K. Singh, C.-N. Lee. "A Multiple Controlled Toffoli Driven Adaptive Quantum Neural Network Model for Dynamic Workload Prediction in Cloud Environments." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024. doi:10.1109/TPAMI.2024.3402061
2. J. Kumar, D. Saxena, K. Gupta, S. Kumar, A. K. Singh. "A Comprehensively Adaptive Architectural Optimization-Ingrained Quantum Neural Network Model for Cloud Workloads Prediction." *IEEE Transactions on Neural Networks and Learning Systems*, 2025. doi:10.1109/TNNLS.2025.3577721
3. H. Yuan, J. Bi, S. Li, J. Zhang, M. Zhou. "An Improved LSTM-Based Prediction Approach for Resources and Workload in Large-Scale Data Centers." *IEEE Internet of Things Journal*, 2024. doi:10.1109/JIOT.2024.3383512
4. F. Zhao, W. Lin, S. Lin, H. Zhong, K. Li. "TFEGRU: Time-Frequency Enhanced Gated Recurrent Unit With Attention for Cloud Workload Prediction." *IEEE Transactions on Services Computing*, 2024. doi:10.1109/TSC.2024.3517324
5. F. Zhao, W. Lin, S. Lin, S. Tang, K. Li. "MSCNet: Multi-Scale Network With Convolutions for Long-Term Cloud Workload Prediction." *IEEE Transactions on Services Computing*, 2025. doi:10.1109/TSC.2025.3536313
6. J. Li, J. Yao, D. Xiao, D. Yang, W. Wu. "EvoGWP: Predicting Long-Term Changes in Cloud Workloads Using Deep Graph-Evolution Learning." *IEEE Transactions on Parallel and Distributed Systems*, 2024. doi:10.1109/TPDS.2024.3357715
7. Y. Luo, Z. Yu, J. Zheng, Z. Yang, L. Sun, D. Wu. "Causality-Aware Graph Mixture of Experts for Accurate Multi-Cloud Workload Prediction." *IEEE Transactions on Cloud Computing*, 2026. doi:10.1109/TCC.2026.3689897
8. M. Sallam, K. Kaur. "Optimized Resource Forecasting for Carbon-Intelligent Data Centers With TempoSight: A Hybrid Deep Learning Approach." *IEEE Internet of Things Journal*, 2025. doi:10.1109/JIOT.2025.3610895
9. Kollu et al. "Comparative analysis of cloud resources forecasting using deep learning techniques based on VM workload traces." *Transactions on Emerging Telecommunications Technologies*, 2024. doi:10.1002/ett.4933
10. N. K. Biswas, D. Das, S. Banerjee, U. Biswas. "Energy-Aware VM Consolidation Using Similarity-Driven Intelligence in Green Cloud Environments." *IEEE Transactions on Sustainable Computing*, 2026. doi:10.1109/TSUSC.2026.3660014
11. F. Meng, H. Dai, G. Cong, B. Zhu, H. Zhao. "CATScaler: A Convolution-Augmented Transformer Scaling Framework for Cloud-Native Applications." *IEEE Transactions on Services Computing*, 2025. doi:10.1109/TSC.2025.3592383
12. Y. Kim, S. Song, B.-M. Koo, J. Son, Y. Lee, J.-G. Baek. "Enhancing Long-Term Cloud Workload Forecasting Framework: Anomaly Handling and Ensemble Learning in Multivariate Time Series." *IEEE Transactions on Cloud Computing*, 2024. doi:10.1109/TCC.2024.3400859

13. S. Lin, W. Lin, F. Zhao, H. Chen. "Benchmarking and revisiting time series forecasting methods in cloud workload prediction." *Cluster Computing*, 2024. doi:10.1007/s10586-024-04827-3

Dataset

S. Shen, V. van Beek, A. Iosup. "Statistical Characterization of Business-Critical Workloads Hosted in Cloud Datacenters." *CCGrid* 2015. GWA-T-12 Bitbrains, fastStorage trace. Data courtesy of Bitbrains IT Services, published by TU Delft.